

Speech and Audio Processing

06 - self-supervised learning

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Class schedule (tentative)

01	Oct. 03	Introduction	Learn from basic to state-of-the-art about speech processing.
02	Oct. 10	Speech production / 音声生成	
03	Oct. 17	Speech synthesis / 音声合成	
04	Oct. 24	Speech synthesis 2 / 音声合成2	
05	Oct. 31	Speech recognition / 音声認識	
06	Nov. 7	Self-supervised learning / 自己教師あり学習	
07	Nov. 14	Sound recognition/synthesis / 環境音認識合成	
08	Nov. 28	Foundation model / 基盤モデル	
09	Dec. 05	Advanced topics / 発展的課題	
10	Dec. 12	Speech system building / 音システム構築	Learn how to build your speech system.
11	Dec. 19	Speech system building / 音システム構築	
12	Dec. 26	Speech system building / 音システム構築	
13	Jan. 09	Speech system building / 音システム構築	
14	Jan. 23	Final presentation / 最終発表	

Previous class: automatic recognition (speech to text)

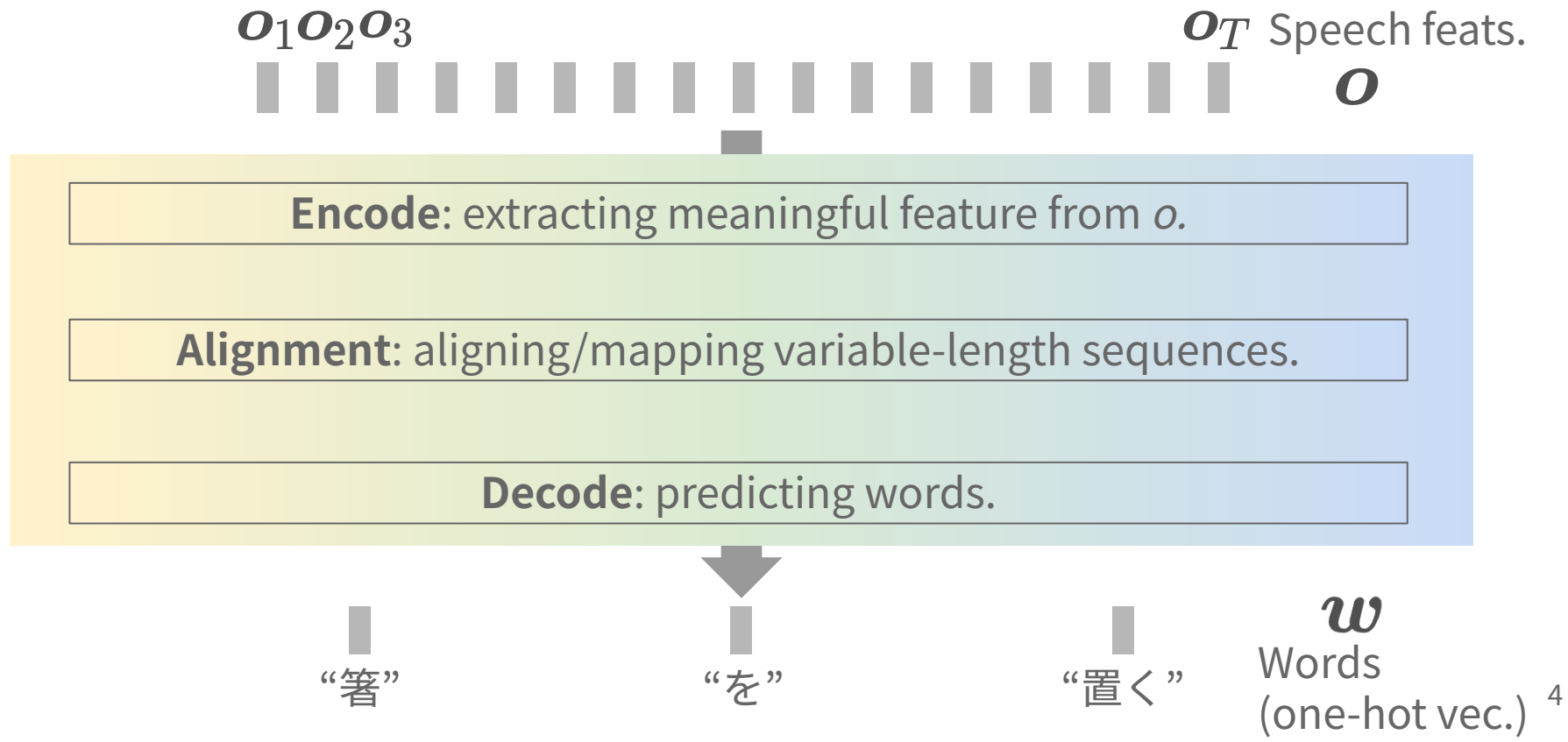
Mapping speech into text.

**Expanding
speech
recognition**

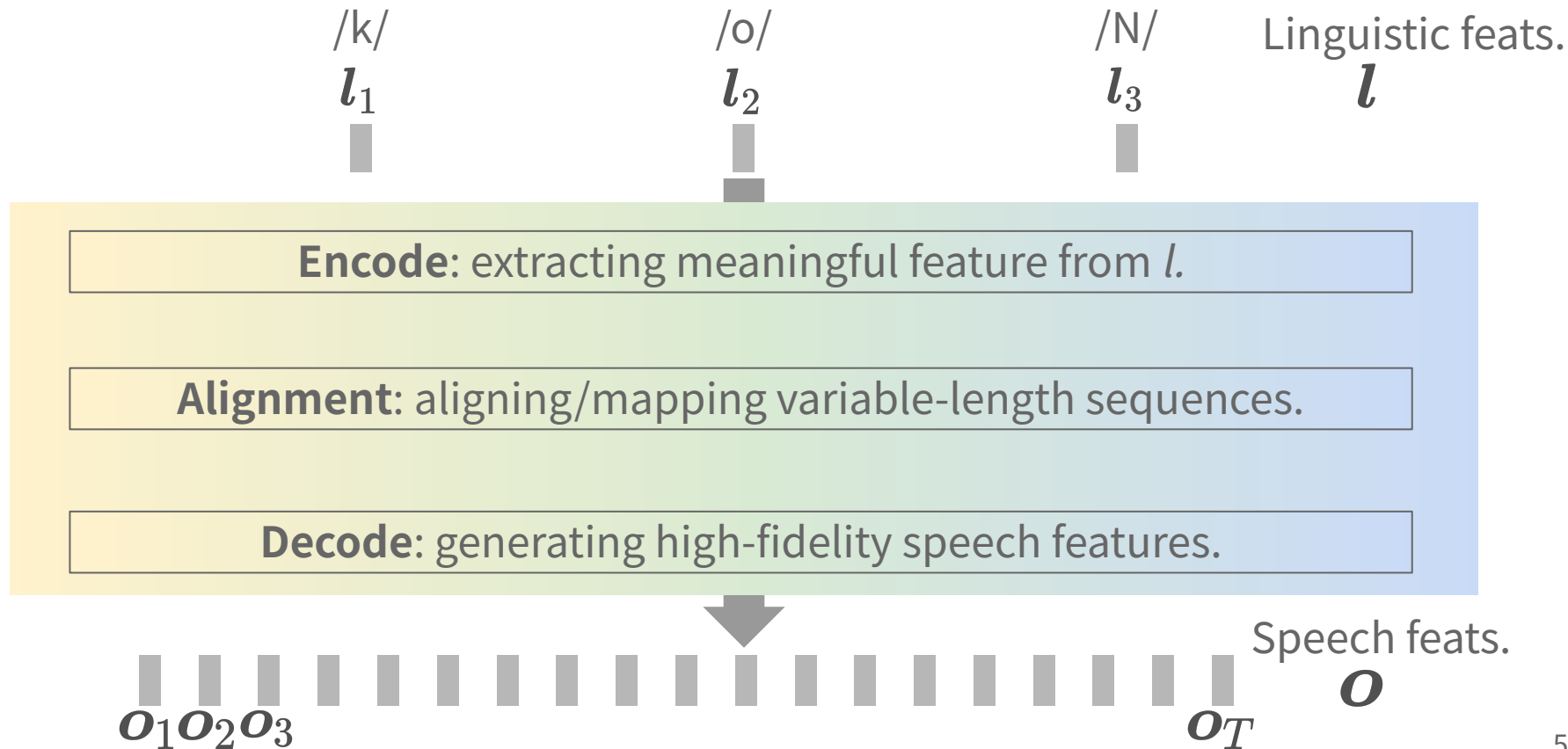
Google Research



Three inner modules in **speech-to-text**



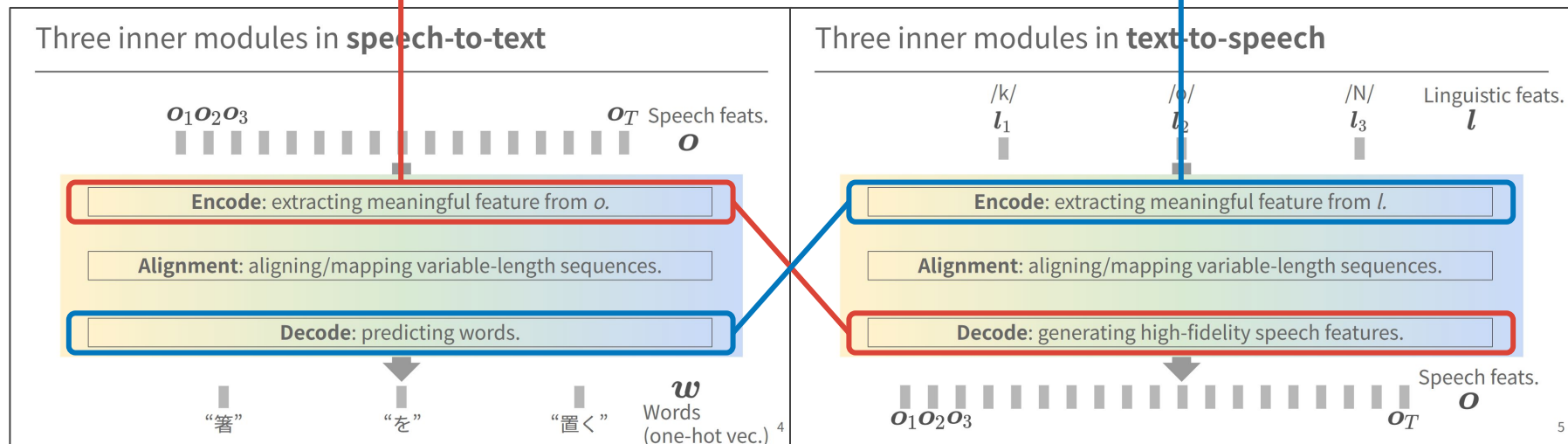
Three inner modules in **text-to-speech**



What encoders and decoders learn

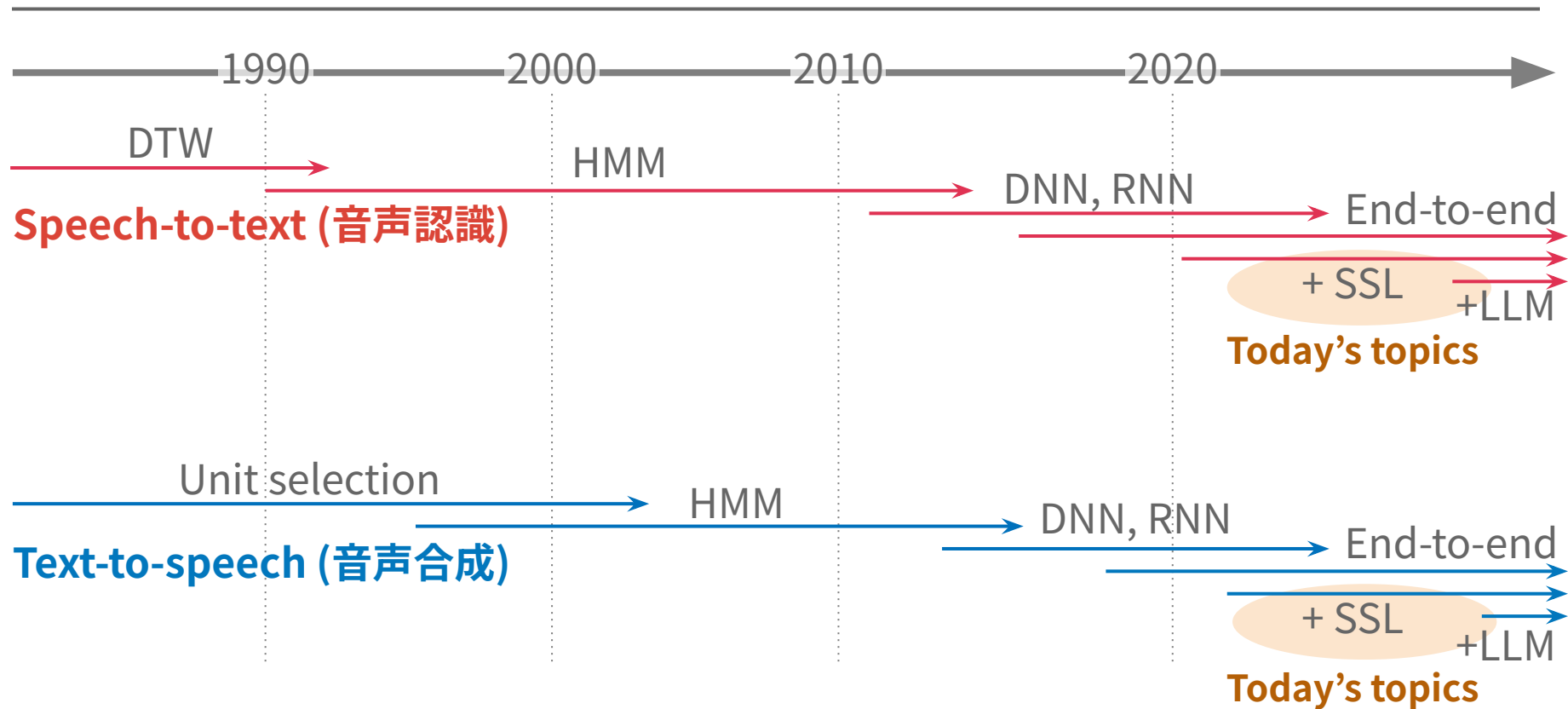
Learn to understand/generate **speech**.
no text

Learn to understand/generate **text**.
no speech



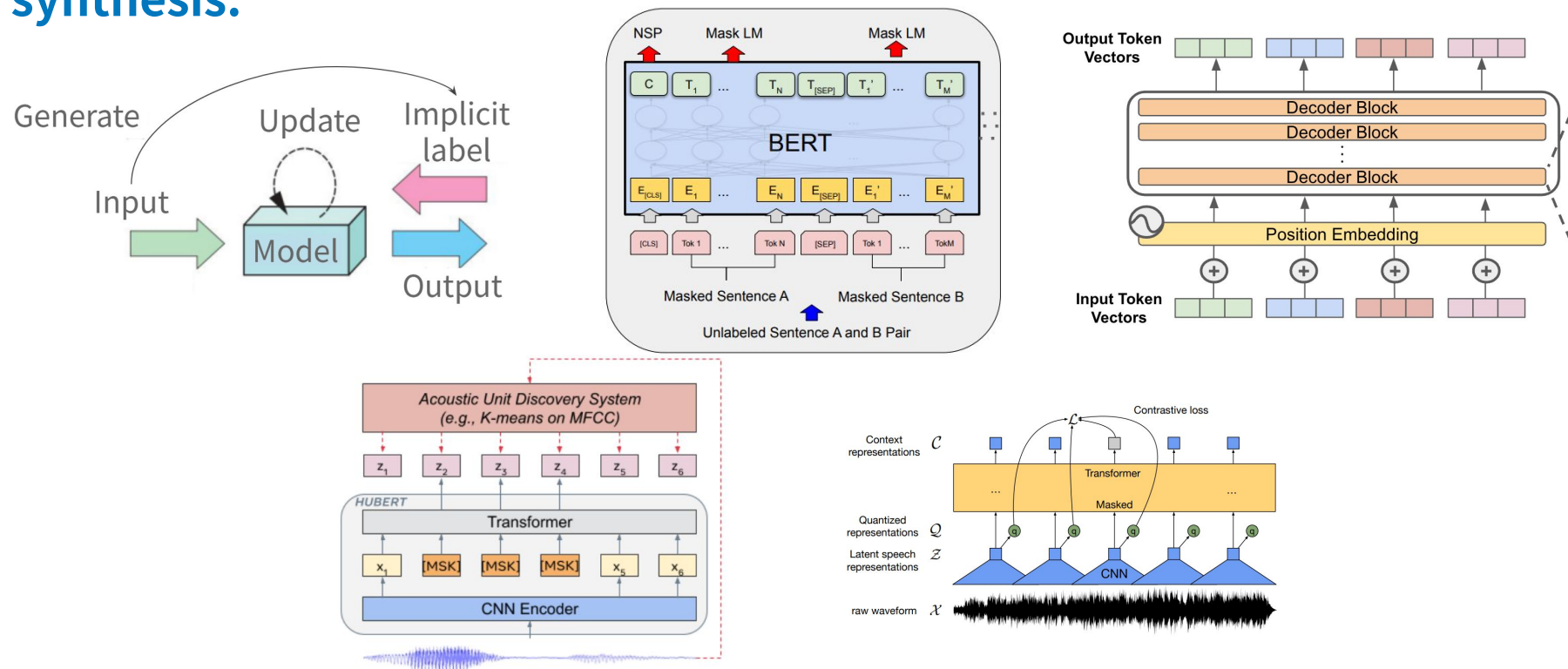
Can we **pre-train** models on **speech-only** or **text-only data**?
easier to collect than speech-text pairs

History



Today's goal: self-supervised learning

Pretraining models on unimodal data for speech recognition and synthesis.

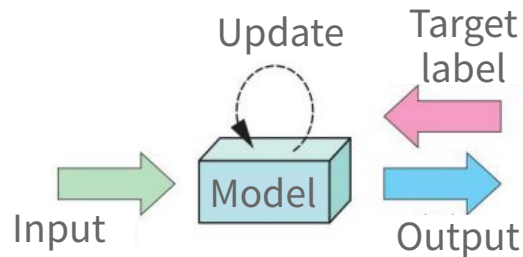


What is self-supervised learning (SSL)?
/ 自己教師あり学習って何？

Several types of machine learning

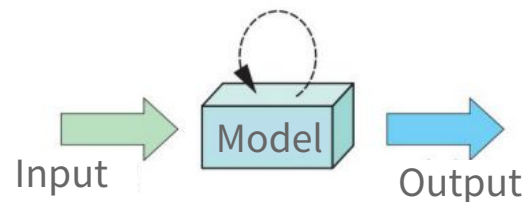
- **Supervised learning / 教師あり学習**

- Using labeled data
- Regression, classification, etc.



- **Unsupervised learning on unlabeled data / 教師なし学習**

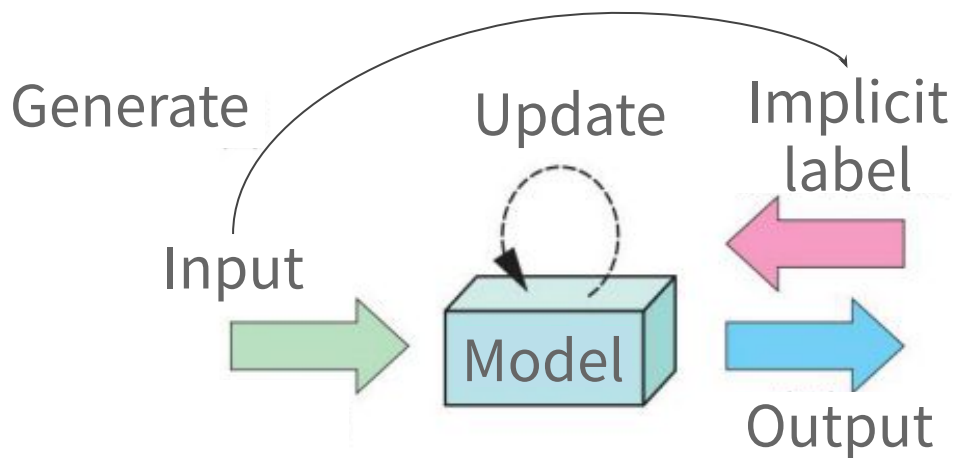
- Using unlabeled data
- Clustering, representation learning, etc.



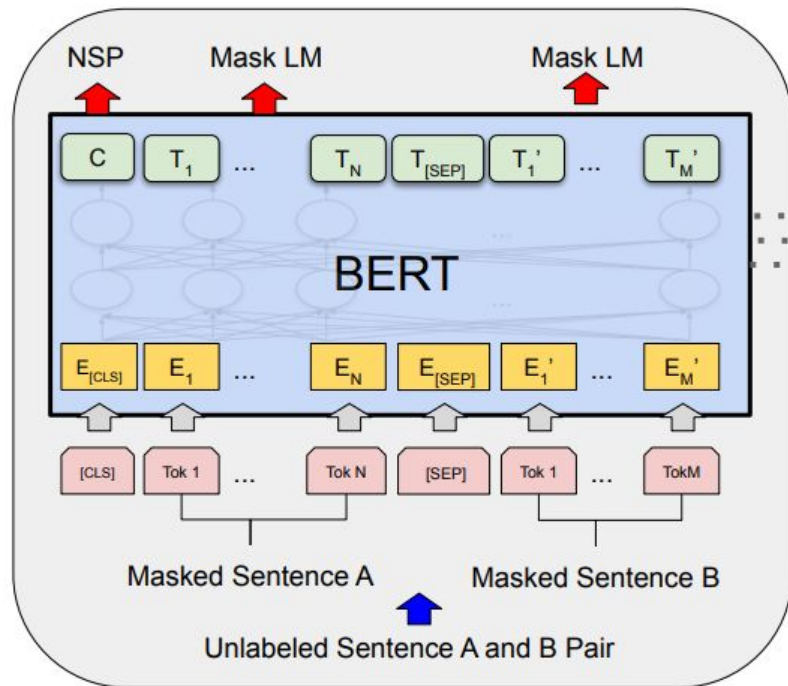
- Weakly-supervised learning / 弱教師あり学習
- Semi-supervised learning / 半教師あり学習
- Reinforcement learning / 強化学習
- Meta learning / メタ学習

Self-supervised learning / 自己教師あり学習

- “Self-supervised learning is a machine learning technique that uses unsupervised learning (...). Rather than relying on labeled datasets for supervisory signals, self-supervised models generate implicit labels from unstructured data.”
/ 自己教師あり学習とは、(中略) 教師なし学習を用いる機械学習手法であり、ラベル付きデータではなく、ラベル無しデータから生成した暗黙的なラベルを利用する。



Self-supervised learning for **text**



BERT (deep **bidirectional** Transformers)

- Transformer-based model
- Training with two objectives

NSP (next sentence prediction)

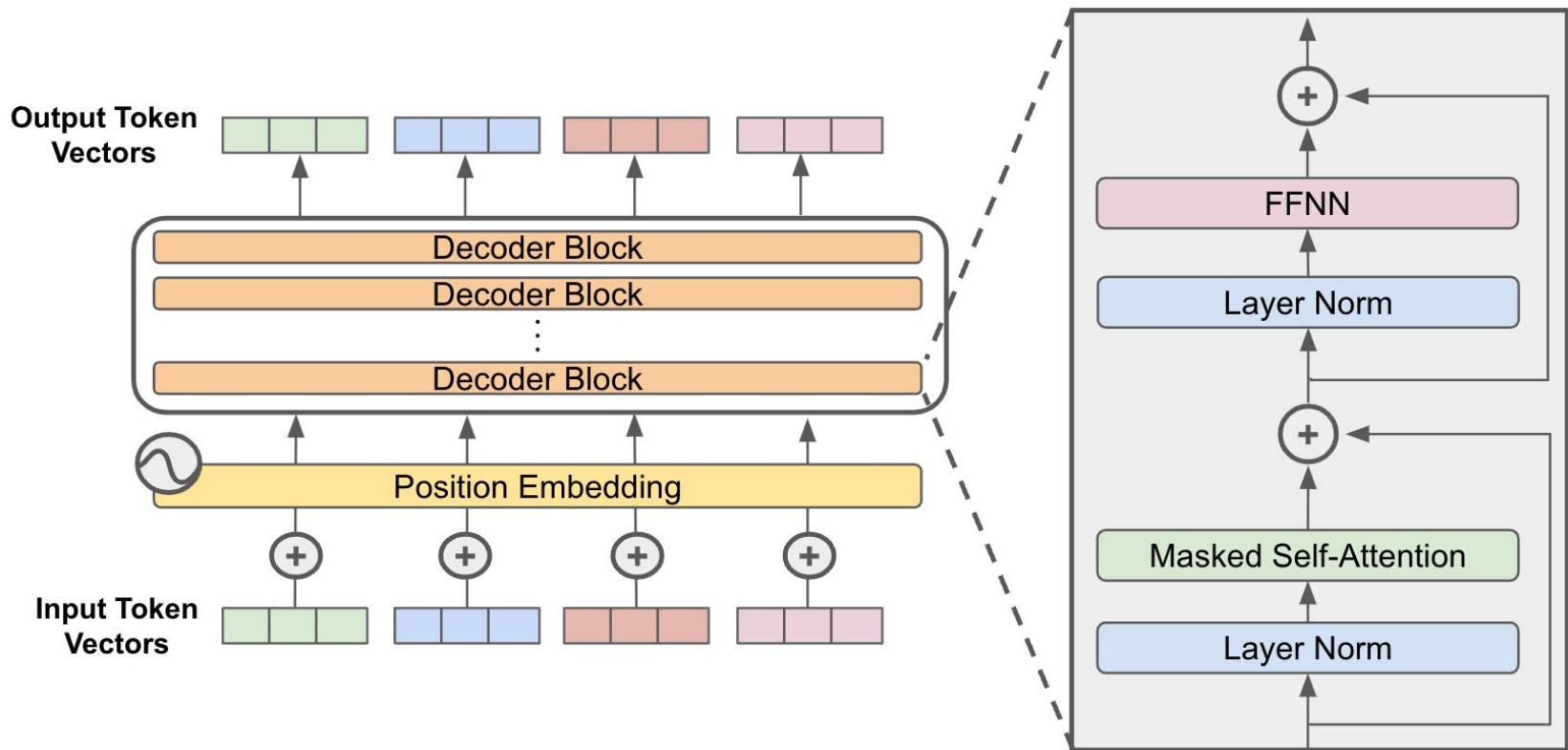
- “True” if the sentence B is actual subsequent of the sentence A.

MLM (masked language model)

- Predicts randomly masked tokens from surround unmasked tokens.

<https://arxiv.org/pdf/1810.04805>

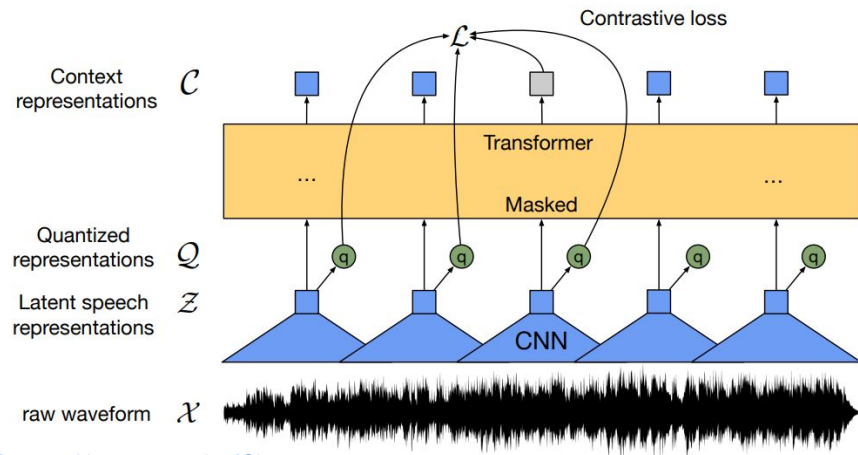
Self-supervised learning for **text**



GPT (generative pre-trained Transformer) is the **decoder-only** Transformer.

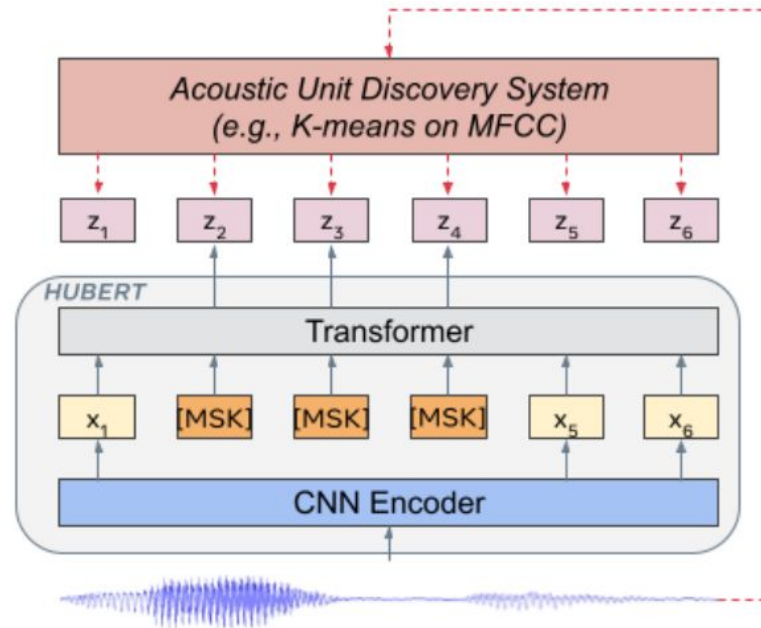
<https://cameronwolfe.substack.com/p/decoder-only-transformers-the-workhorse>

Self-supervised learning for speech (today's topic!)



<https://arxiv.org/pdf/2006.11477>

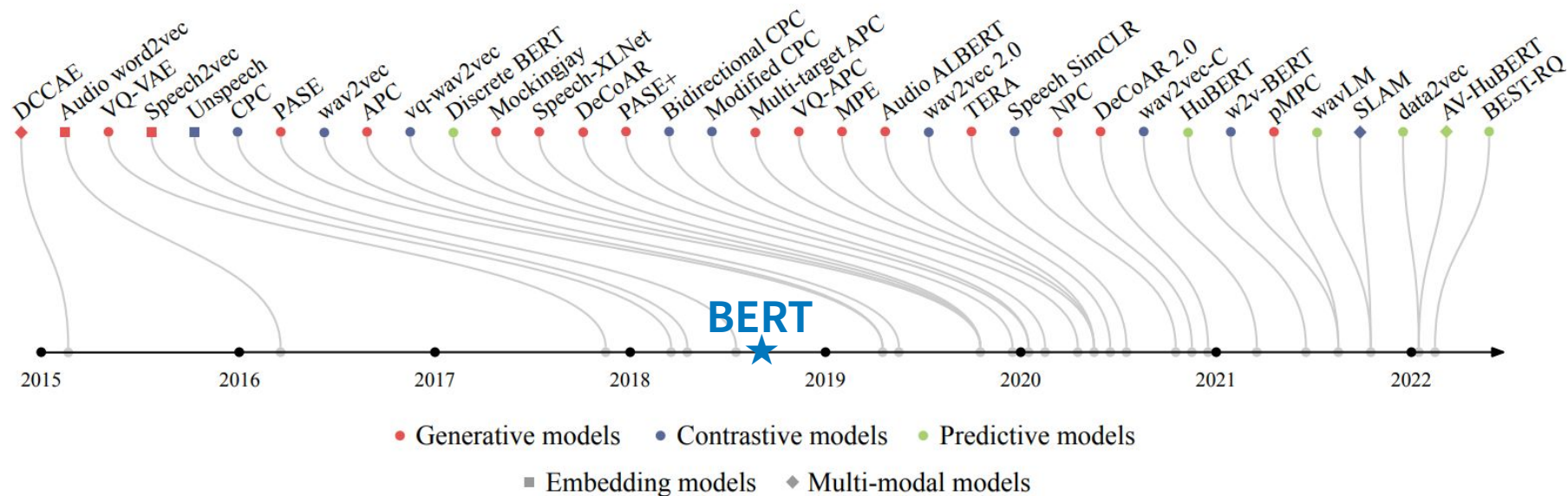
wav2vec 2.0



<https://arxiv.org/pdf/2106.07447>

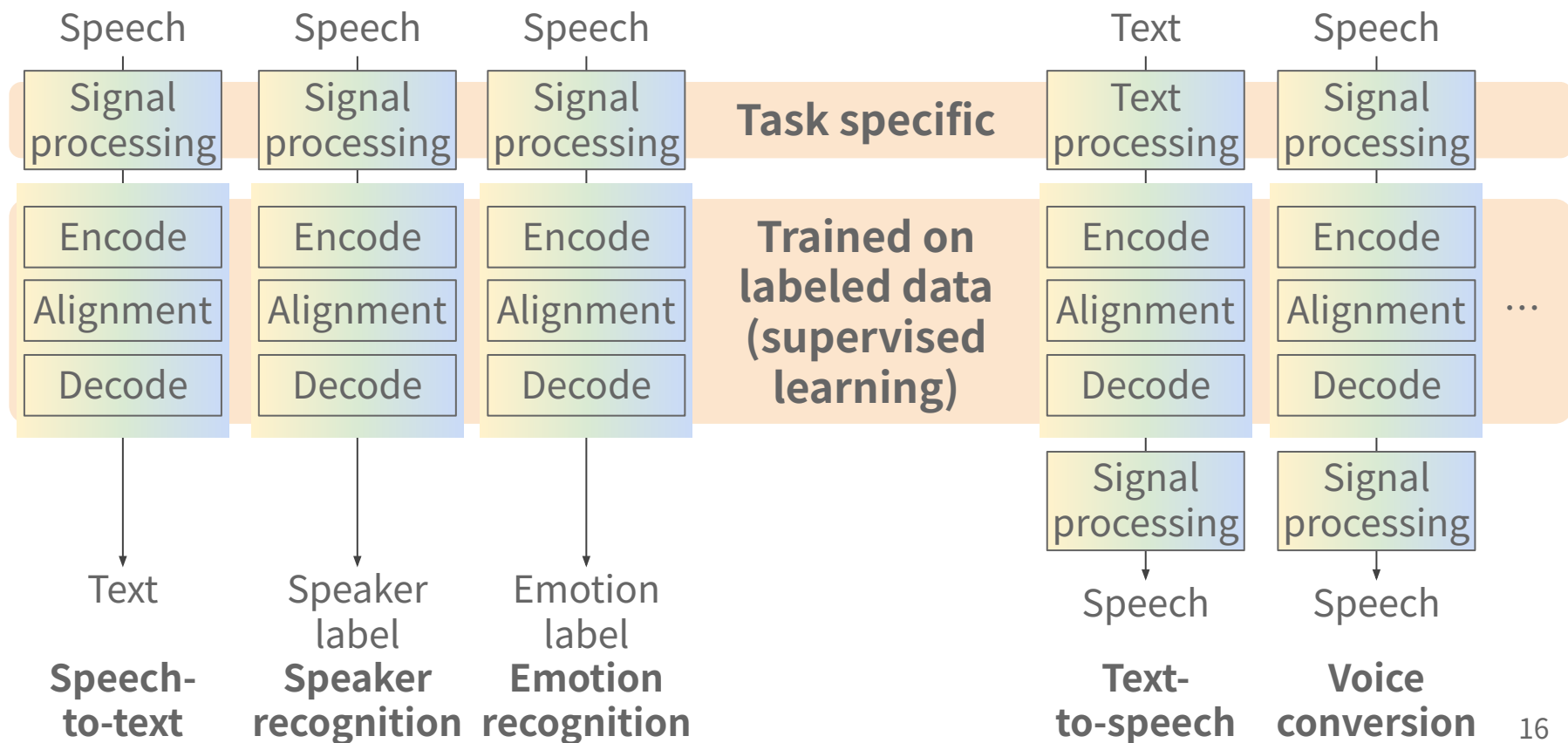
HuBERT

History of self-supervised learning for speech

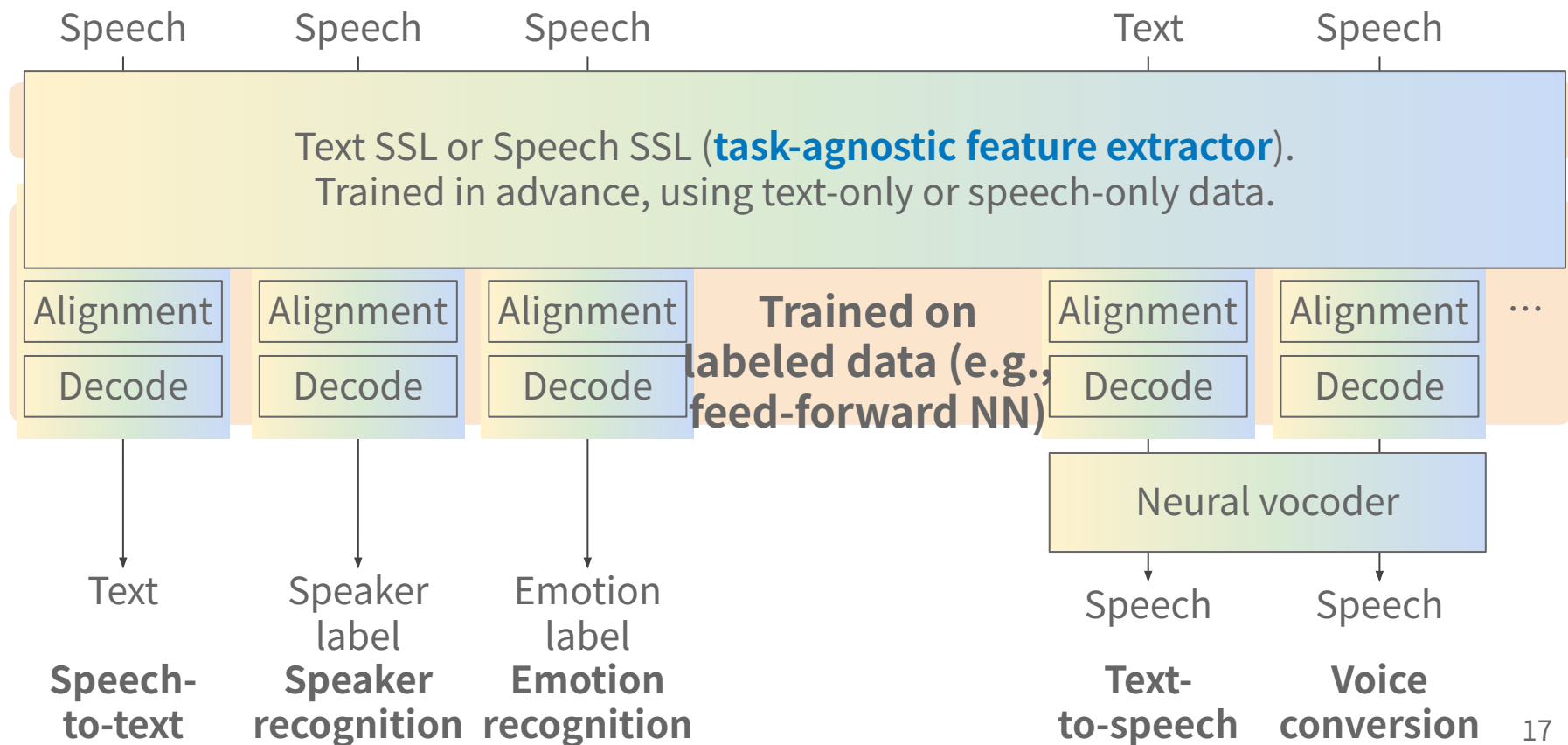


<https://arxiv.org/pdf/2205.10643>

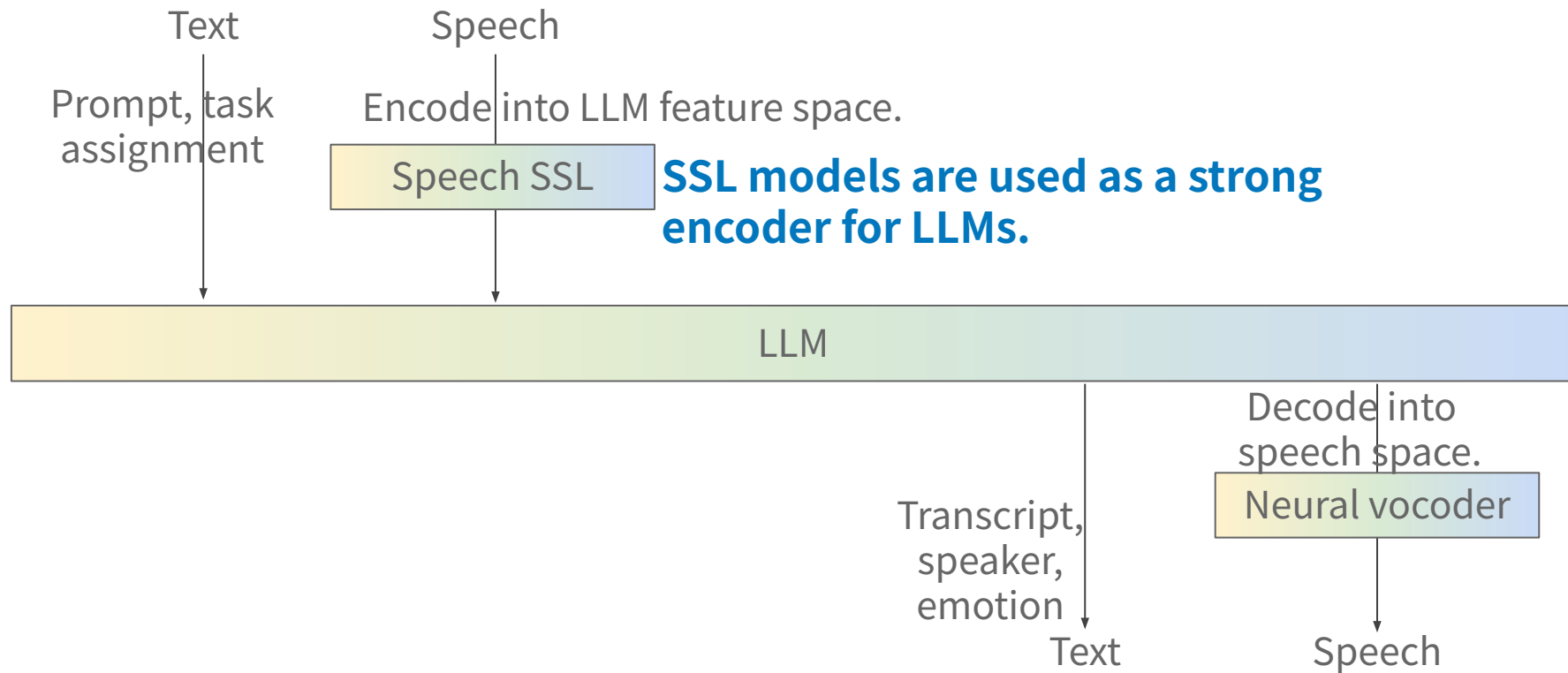
pre-SSL era



SSL era



(...and LLM era)



Formulation of self-supervised learning / 自己教師あり学習の定式化

Three types of self-supervised learning (for speech)

- **Contrastive approaches / 对照型**

- Create positive and negative samples.
- Move the positive sample close and the negative sample far.

- **Predictive approaches / 予測型**

- (Typically) drop features.
- Predict the dropped features from undropped ones.

- **Generative approaches / 生成型**

- Adopt generative models.
- (Re)generate features via the generative models.

wav2vec 2.0 (2020)

Overview

- Randomly mask CNN outputs.
- Quantize the CNN outputs.
- Move Transformer outputs closer to the CNN outputs.

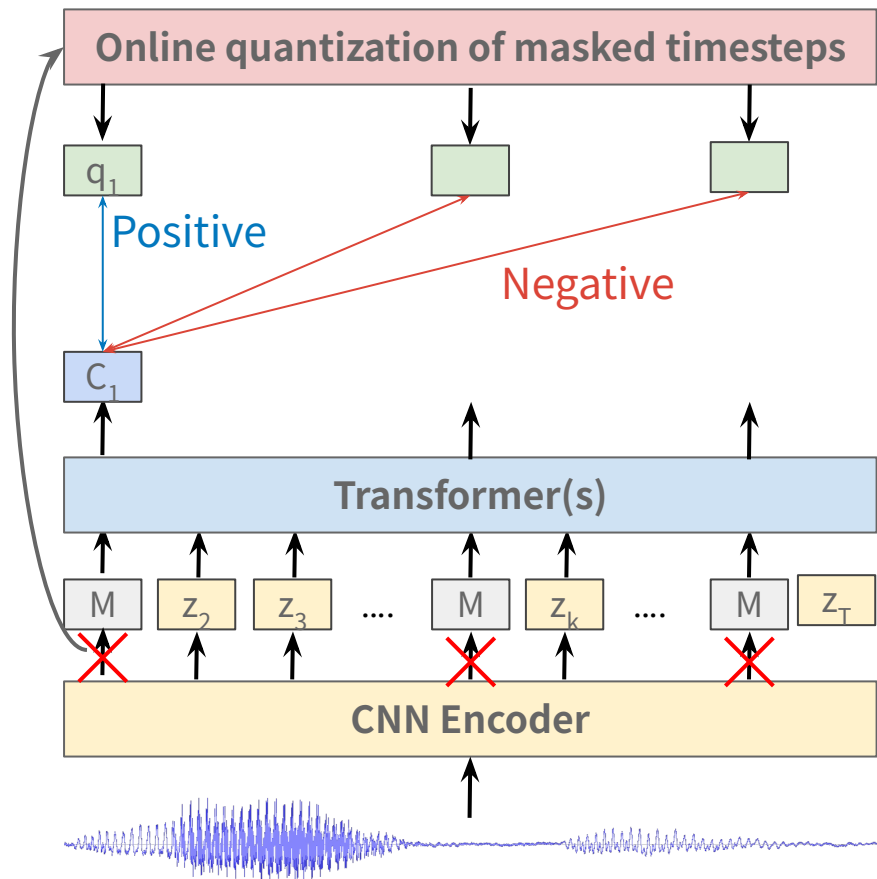
→ **Maximize similarity betw. pos.**

Training objective

Vector similarity to pos
(i.e., move the pos-pair closer)

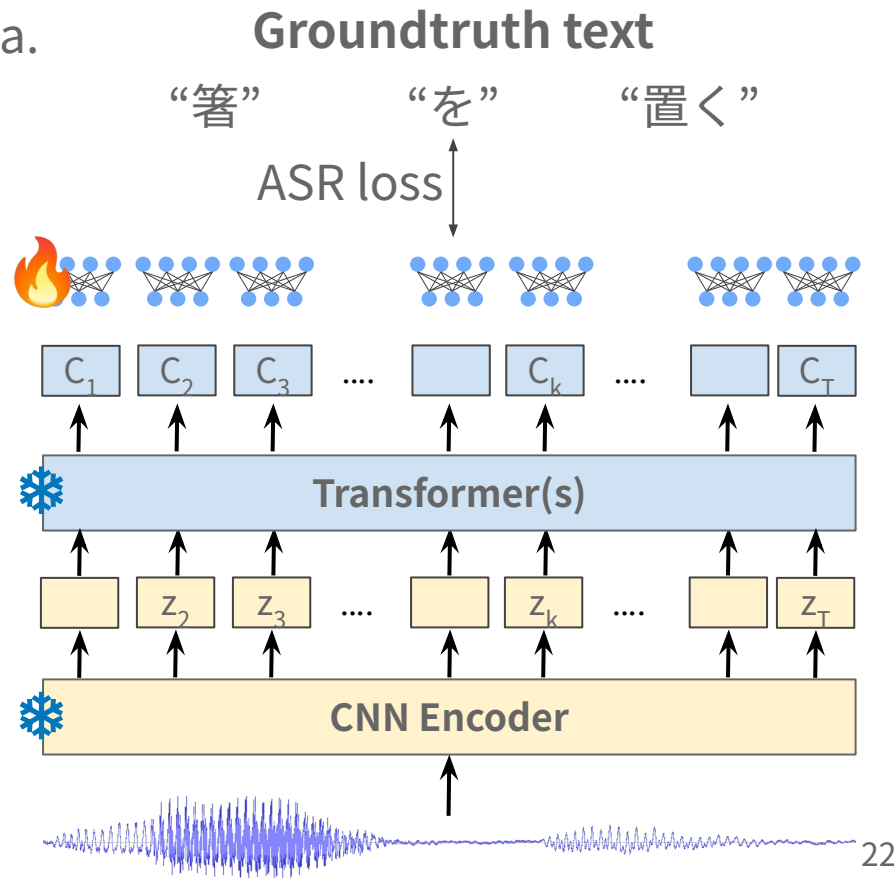
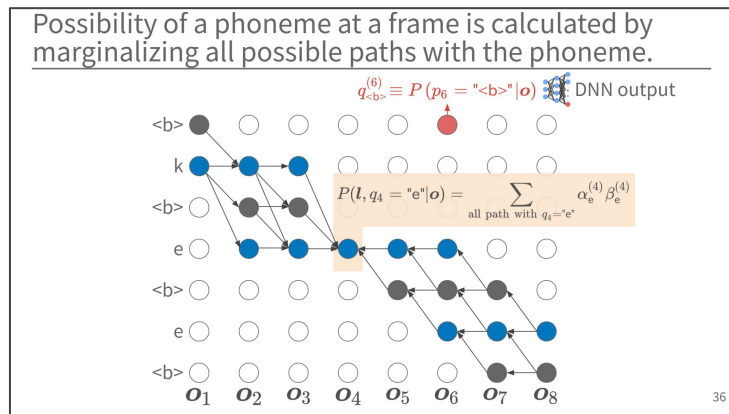
$$\mathcal{L}_m = -\log \frac{\exp(\text{sim}(\mathbf{c}_t, \mathbf{q}_t)/\kappa)}{\sum_{\tilde{\mathbf{q}} \sim \mathbf{Q}_t} \exp(\text{sim}(\mathbf{c}_t, \tilde{\mathbf{q}})/\kappa)}$$

Vector similarities to pos and negs



wav2vec 2.0 → ASR finetuning

- Pretrains wav2vec on speech-only data.
- Adds a linear layer on top of wav2vec.
-  Trains the linear layer while freezing  wav2vec2 on speech-text pair data.
 - e.g., CTC loss

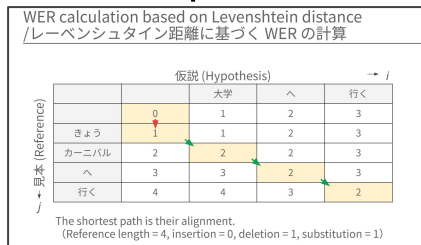
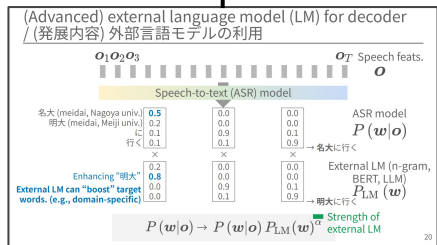


Surprisingly low WER using only 10 min. of pair data.

Table 1: WER on the Librispeech dev/test sets when training on the Libri-light low-resource labeled data setups of 10 min, 1 hour, 10 hours and the clean 100h subset of Librispeech. Models use either the audio of Librispeech (LS-960) or the larger LibriVox (LV-60k) as unlabeled data. We consider two model sizes: BASE (95m parameters) and LARGE (317m parameters). Prior work used 860 unlabeled hours (LS-860) but the total with labeled data is 960 hours and comparable to our setup.

Model	Unlabeled data	LM	dev		test	
			clean	other	clean	other
<u>10 min labeled</u>						
Discrete BERT [4]	LS-960	4-gram	15.7	24.1	16.3	25.2
BASE	LS-960	4-gram	8.9	15.7	9.1	15.6
		Transf.	6.6	13.2	6.9	12.9
LARGE	LS-960	Transf.	6.6	10.6	6.8	10.8
	LV-60k	Transf.	4.6	7.9	4.8	8.2

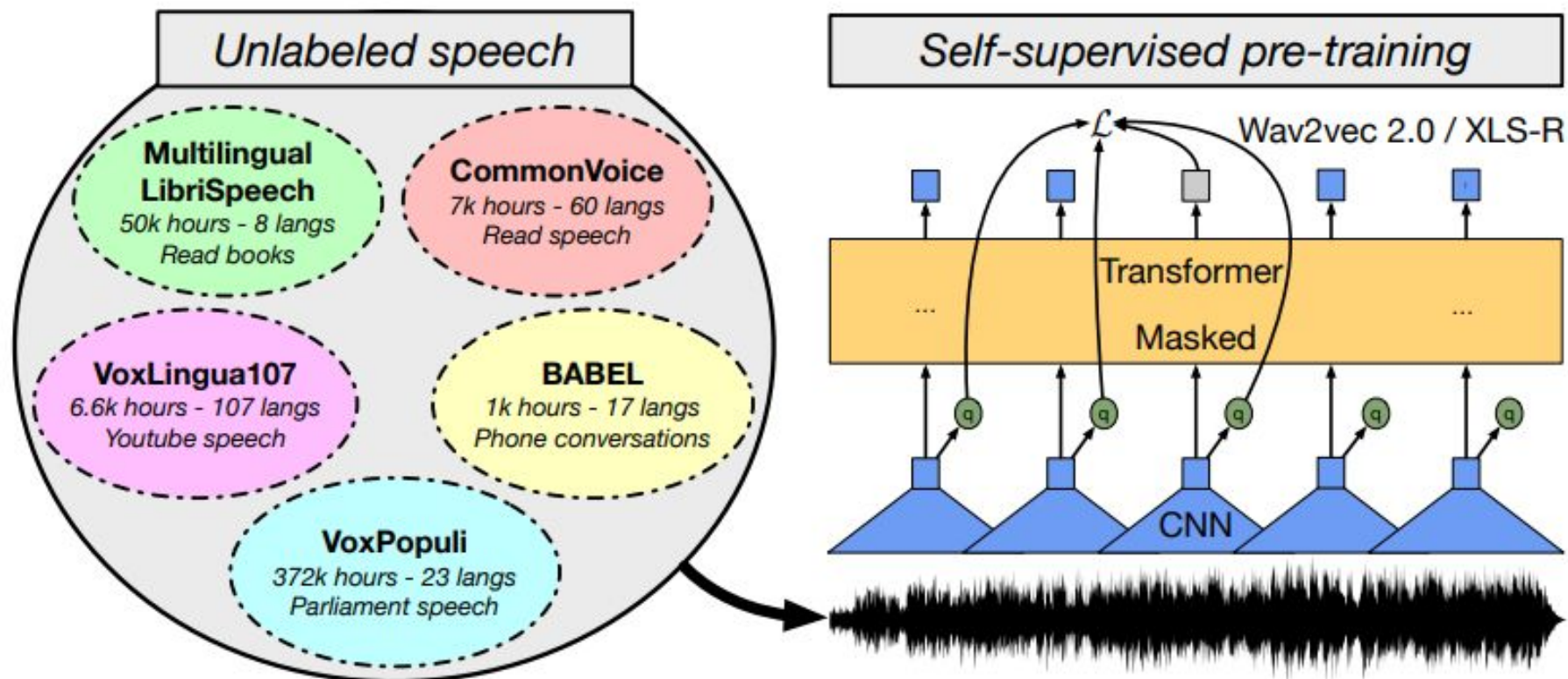
External language model



WER

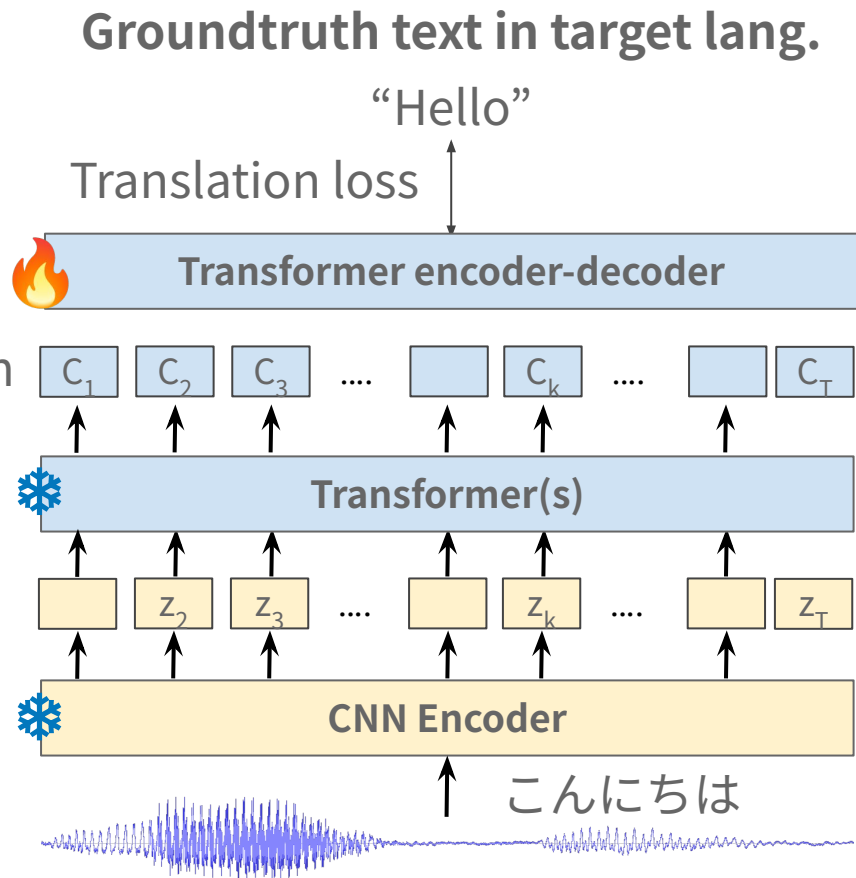
Extending to multi-linguality: XLS-R covering 128 languages

Contrastive / 对照型



XLS-R → speech-to-text translation

- Pretrains XLS-R on multi-lingual speech-only data.
 - The pretrained model is expected to support multilinguality.
- Adds Transformer encoder-decoder on top of XLS-R.
 - Encoder includes self-attn.
 - Decoder includes cross-attn.
-  Trains the Transformer enc-dec.
 - e.g., cross-entropy.



X-Speech to English-text translation

Table 3: Speech translation: results for $X \rightarrow \text{English}$ directions on CoVoST-2 in terms of average BLEU for 21 directions grouped into high/mid/low-resource labeled data directions. Appendix A lists detailed results for all languages.

	high	mid	low	Avg.
<i>Prior work</i>				
XLSR-53 (Conneau et al., 2021)	30.3	11.1	3.2	10.3
VP-100K (Wang et al., 2021b)	27.7	13.2	4.6	11.1
XMEF-En (Li et al., 2021b)	32.4	16.8	4.0	12.4
XMEF-X (Li et al., 2021b)	34.2	20.2	5.9	14.7
<i>This work</i>				
XLS-R (0.3B)	30.6	18.9	5.1	13.2
XLS-R (1B)	34.3	25.5	11.7	19.3
XLS-R (2B)	36.1	27.7	15.1	22.1

Multilingual training significantly contributes to multilingual training.

HuBERT (2021)

Overview

- Quantize speech feats.
- Randomly mask CNN outputs.
- Predict the quantization class at masking frames.

→ Learn to represent “semantic” information from speech.

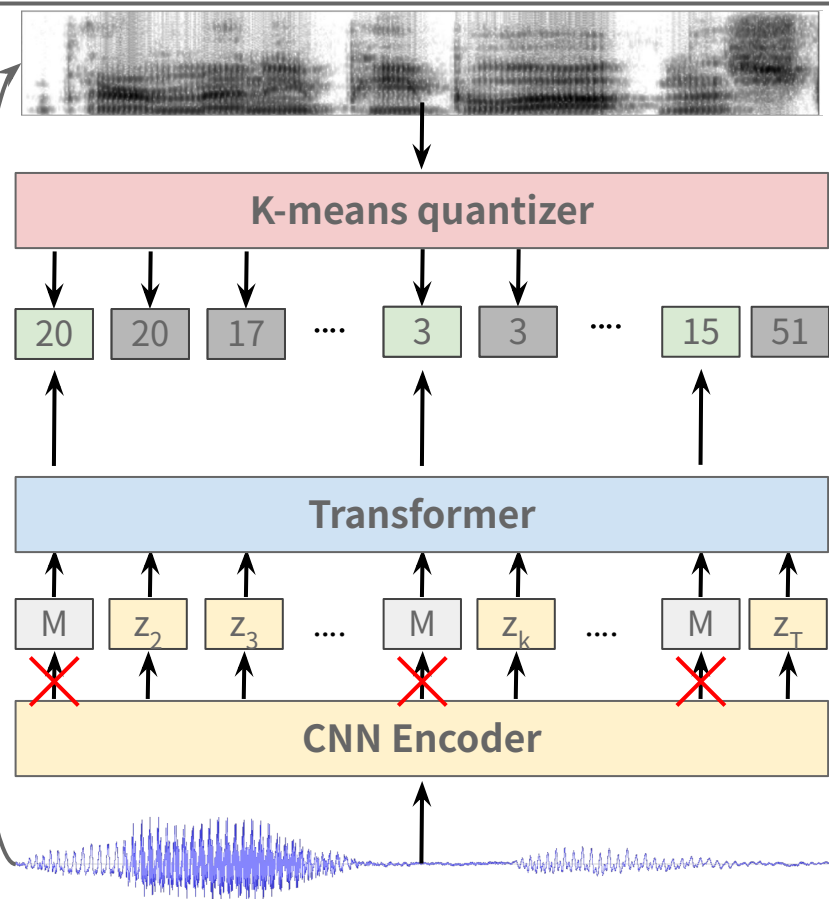
Training objective

Softmax cross entropy

$$\mathcal{L}_m = \sum_{t \in M} -\log p(y_t | X)$$

For all masked frames

* HuBERT feats. are used in the middle of training steps.

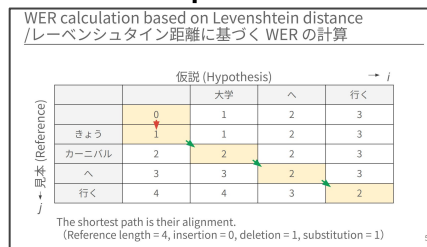
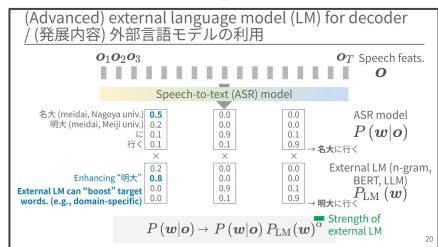


Further improvements than wav2vec 2.0

Model	Unlabeled Data	LM	dev-clean	dev-other	test-clean	test-other
<i>10-min labeled</i>						
DiscreteBERT [51]	LS-960	4-gram	15.7	24.1	16.3	25.2
wav2vec 2.0 BASE [6]	LS-960	4-gram	8.9	15.7	9.1	15.6
wav2vec 2.0 LARGE [6]	LL-60k	4-gram	6.3	9.8	6.6	10.3
wav2vec 2.0 LARGE [6]	LL-60k	Transformer	4.6	7.9	4.8	8.2
HUBERT BASE	LS-960	4-gram	9.1	15.0	9.7	15.3
HUBERT LARGE	LL-60k	4-gram	6.1	9.4	6.6	10.1
HUBERT LARGE	LL-60k	Transformer	4.3	7.0	4.7	7.6
HUBERT X-LARGE	LL-60k	Transformer	4.4	6.1	4.6	6.8

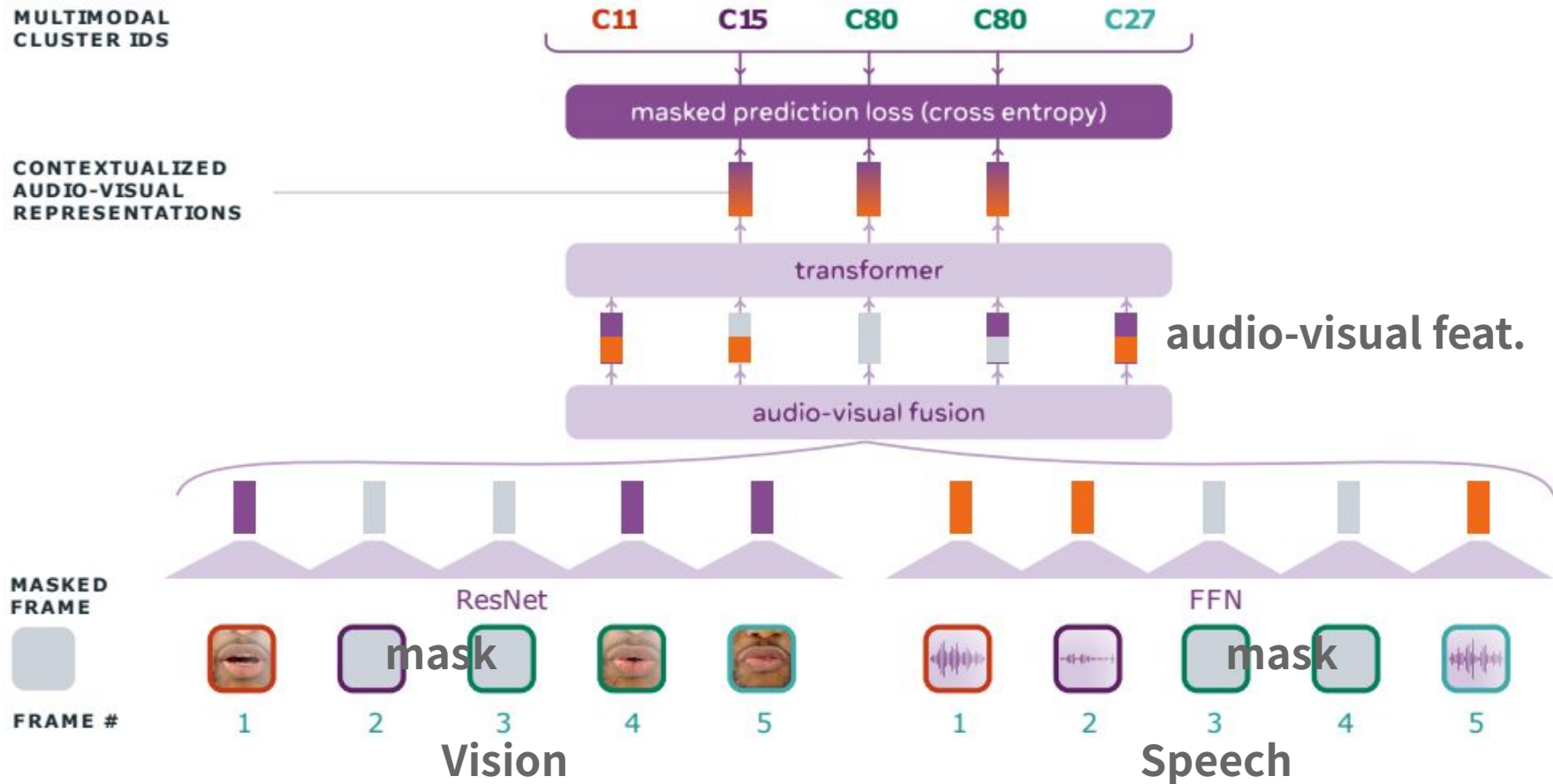
Model size

External language model



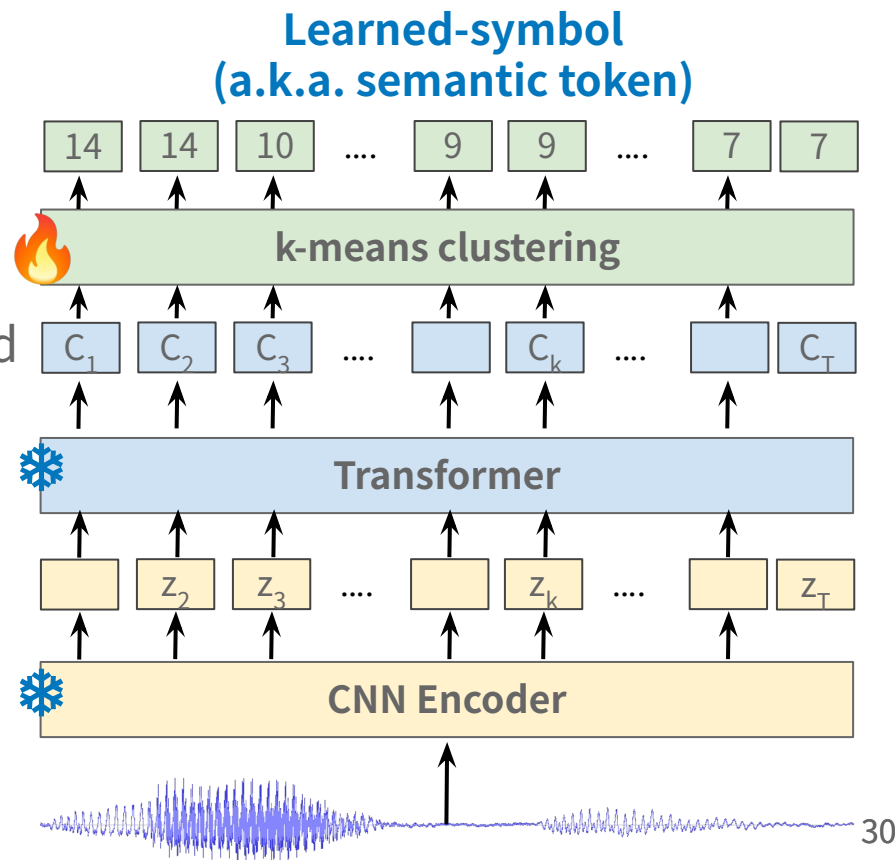
WER

Extending to multi-modal: AV HuBERT supporting speech and vision

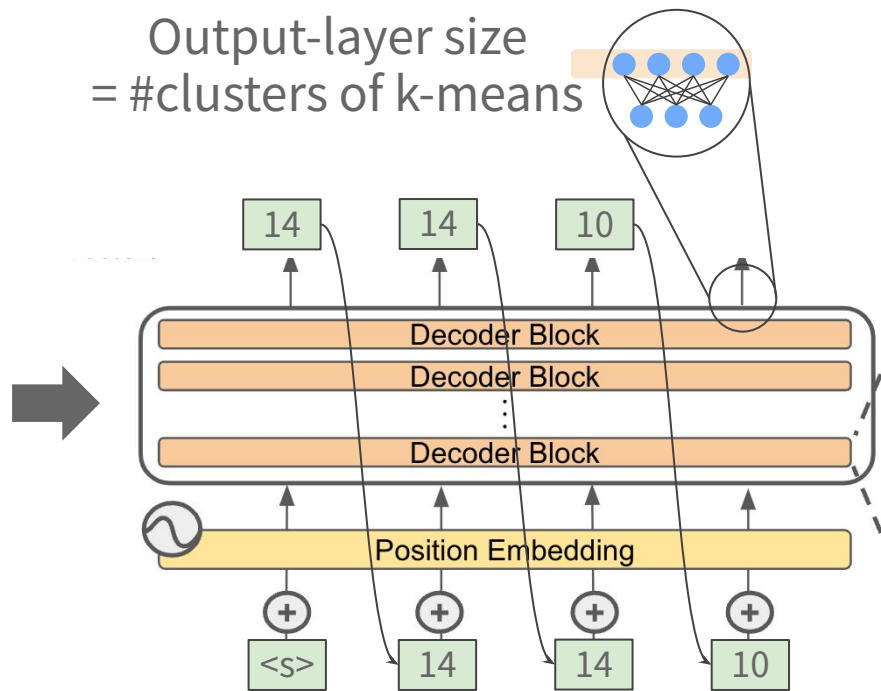
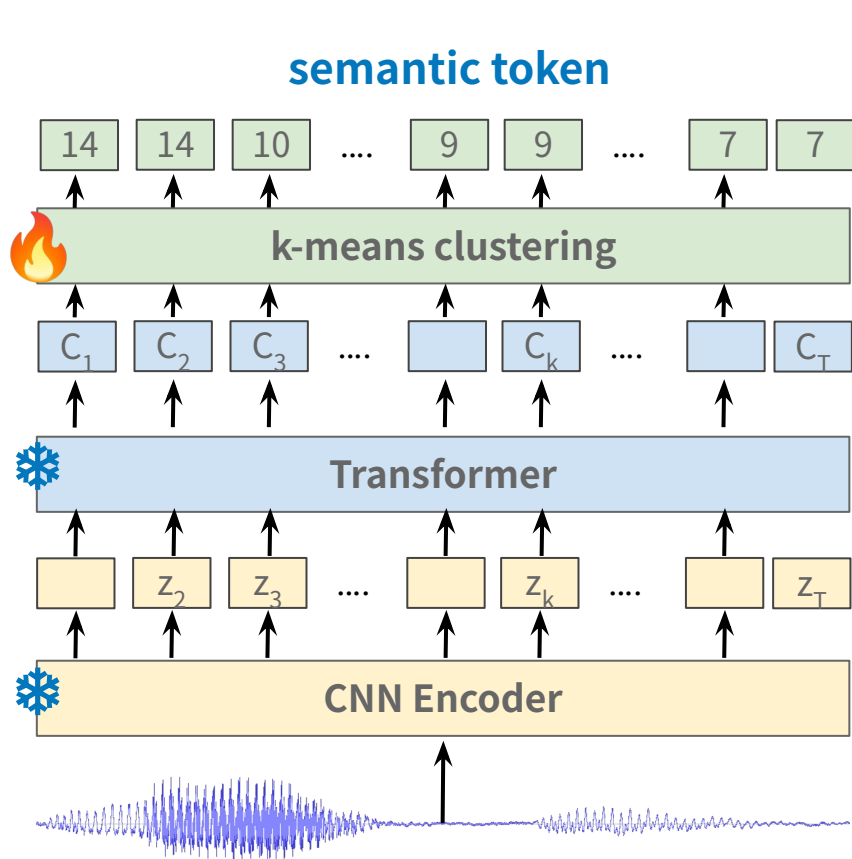


Extending to language: GSLM (generative spoken language modeling)

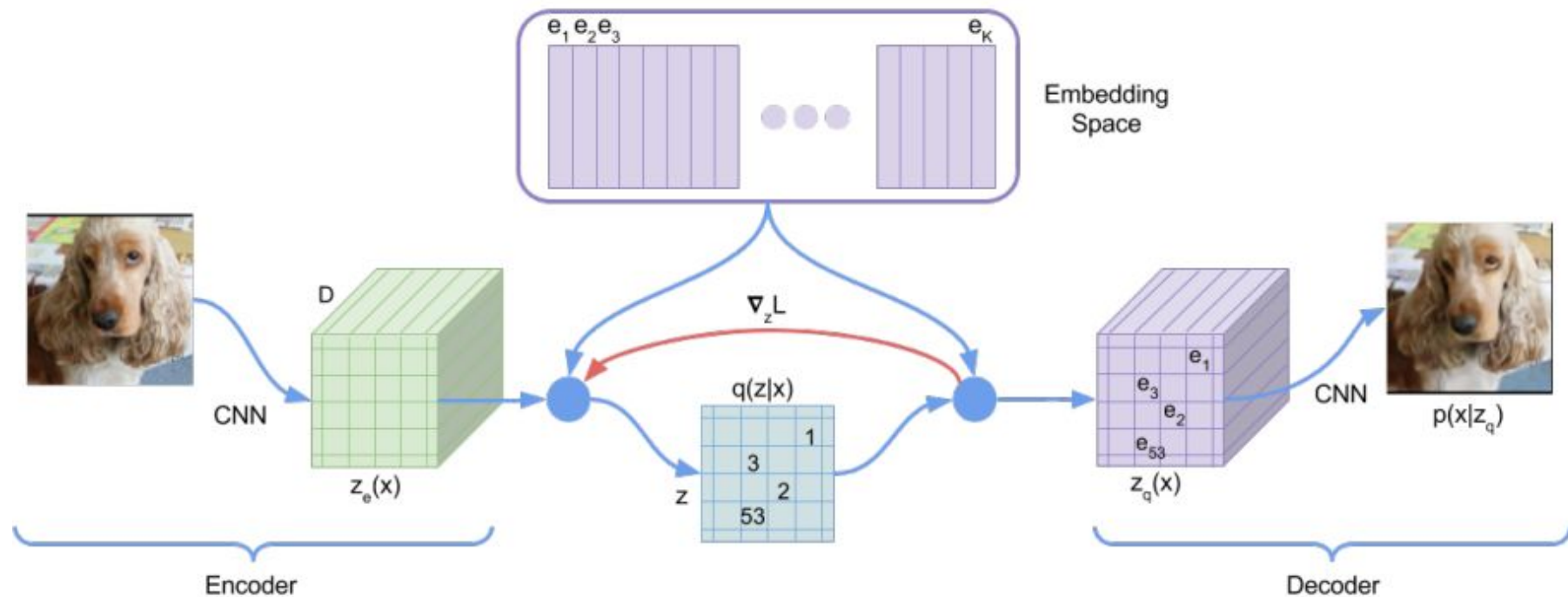
- Quantize the HuBERT outputs (or intermediate-Transformer outputs)
 - They can be seen as pseudo phoneme, mora, or words
- The symbol sequences can be handled like text symbol sequence.
 - e.g., GPT, BERT, etc.



GPT-style modeling of semantic tokens



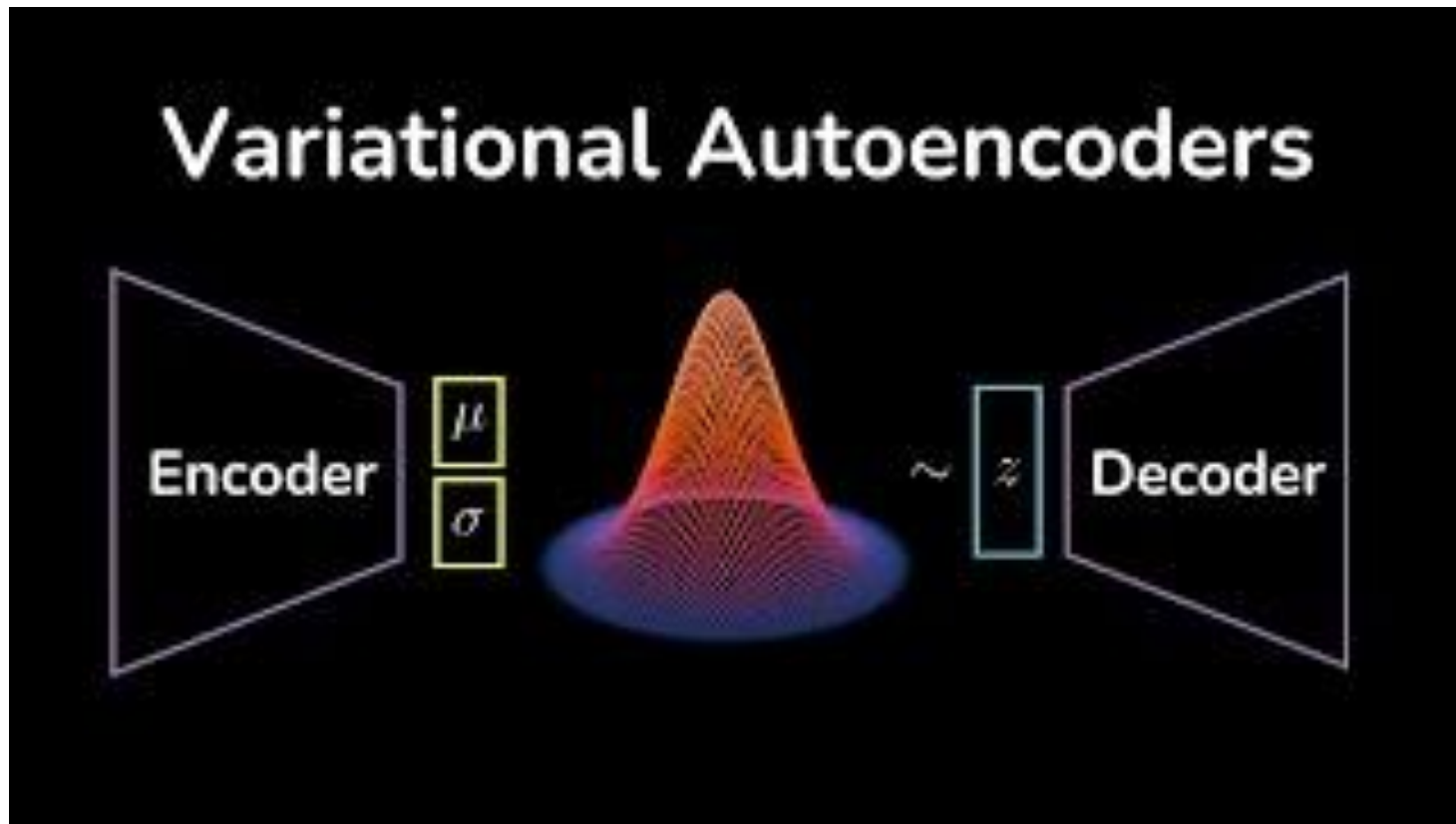
Vector Quantised Variational AutoEncoder (VQ-VAE)



Encoding into quantized vectors.

Decoding from the quantized vectors.

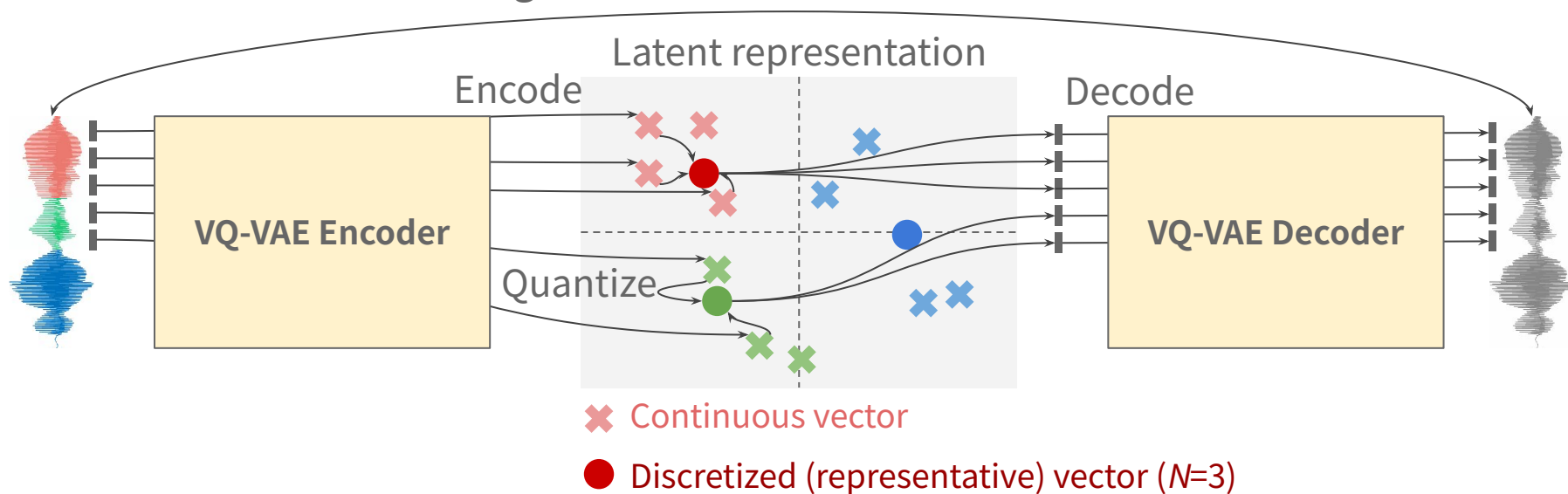
(Revisit) Variational autoencoder



VQ-VAE encodes discretized vectors instead of VAE-like continuous vectors.

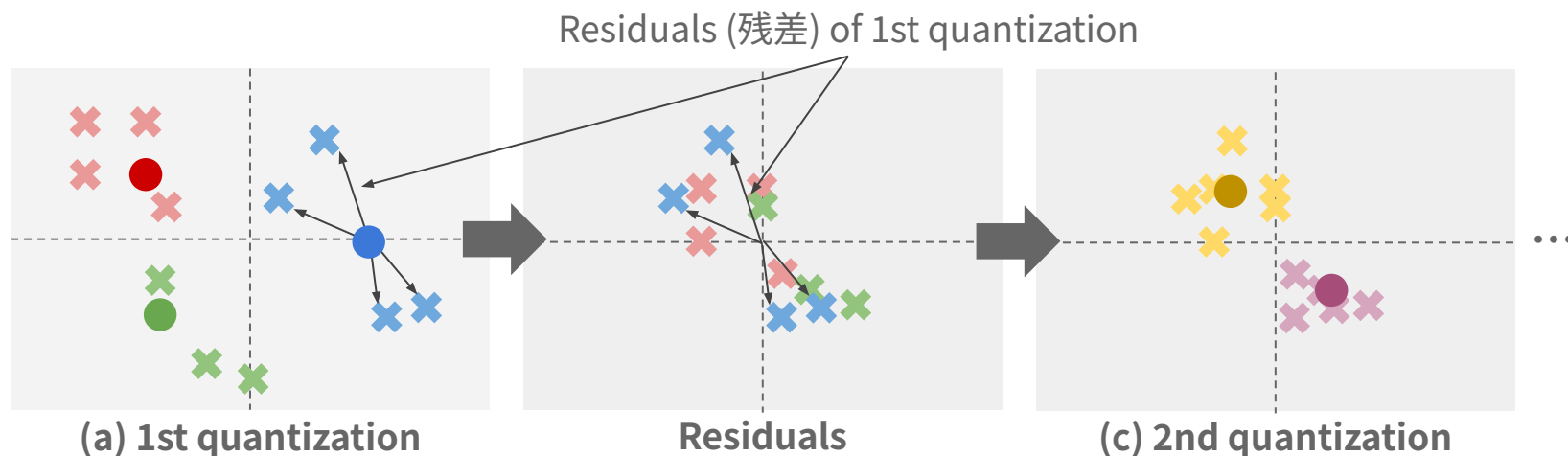
Problem: quantization has modeling errors.

Training encoder & decoder to reconstruct.



Approximation by quantization (量子化) loses detailed information to reconstruct speech waveform. -> Increasing N ? Any smarter way?

Residual vector quantization / 残差ベクトル量子化



Algorithm 1: Residual Vector Quantization

Input: $y = \text{enc}(x)$ the output of the encoder, vector
quantizers Q_i for $i = 1..N_q$

Output: the quantized $\hat{y}$

$\hat{y} \leftarrow 0.0$

residual $\leftarrow y$

for $i = 1$ to N_q **do**

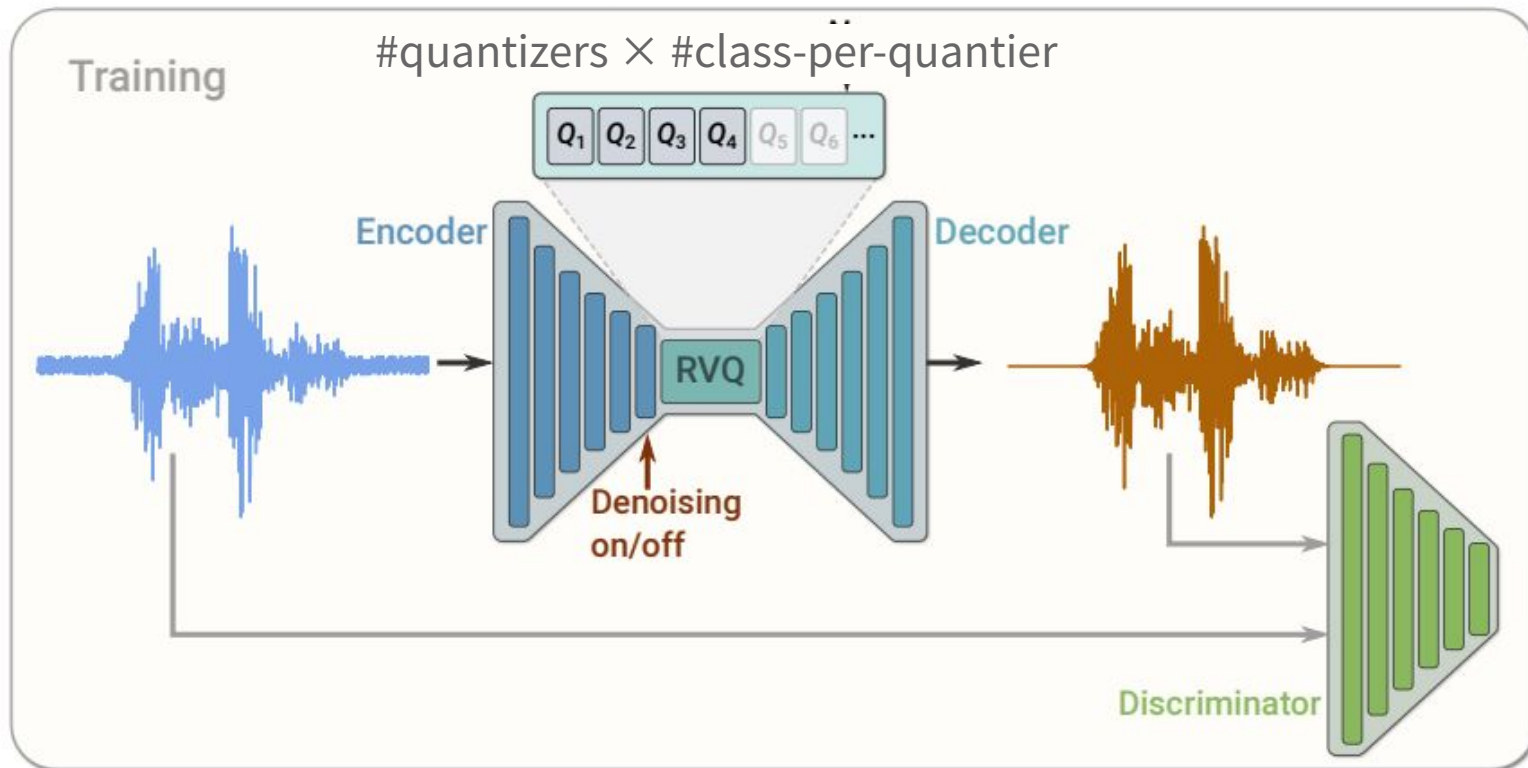
$\hat{y} += Q_i(\text{residual})$

 residual $-= Q_i(\text{residual})$

return $\hat{y}$

SoundStream

a.k.a., acoustic token



Reconstruction via GAN

(Revisit) Three types of self-supervised learning

- **Contrastive approaches / 对照型** ➡ Learns discriminative features.
 - Create positive and negative samples.
 - Move the positive sample close and the negative sample far.
- **Predictive approaches / 予測型** ➡ Learns semantic features. (e.g., semantic tokens)
 - (Typically) drop features.
 - Predict the dropped features from undropped ones.
- **Generative approaches / 生成型** ➡ Learns acoustic features. (e.g., acoustic tokens)
 - Adopt generative models.
 - (Re)generate features via the generative models.

Opensource SSL models / オープンソースの SSL モデル

Speech SSL list (as of 2022)

Model (a)	Input (b)	Corruption (c)	Target (d)
GENERATIVE MODELS			
Audio Word2vec [118], VQ-VAE [75]	X	-	X
Speech2Vec [114], Audio2Vec [116] - skip-gram	$X_{[t_1, t_2]}$	-	$X_{[t_0, t_1]}, X_{[t_2, t_3]}$
Speech2Vec [114], Audio2Vec [116] - cbow	$X_{[t_0, t_1]}, X_{[t_2, t_3]}$	-	$X_{[t_1, t_2]}$
PASE [69], PASE+ [113] ¹	X	-	Different modalities of X
APC [92], [98]	$X_{[1, t]}$	-	$x_{t+c}, c \geq 1$
Speech-XLNet [109]	Xp_t		$x_i \sim \mathcal{P}_i^c$
DeCoAR [99]	$X_{[1, t-1]}, X_{[t+k+1, T]}$	-	$X_{[t, t+k]}$
Mockingjay [100], Audio ALBERT [119], DeCoAR 2.0 [110]	$X_{-[t, t+k]}$		$X_{[t, t+k]}$
TERA [106], BMR [103]	$X_{-[t, t+k]}^{[f, f+j]}$		X
pMPC [105]	$X_{-[t, t+k']} (X_{[t, t+k']}$ is a phoneme)		$X_{[t, t+k']}$
MPE [102]	X	$Z_{-[t, t+k]}$	Z
NPC [107]	X	$Z_{-[t, t+k]}$	X
CONTRASTIVE MODELS			
Unspeech [120]	$X_{[t_1, t_2]}$	-	$X_{[t_0, t_1]}, X_{[t_2, t_3]}$
CPC [56], wav2vec [121], Modified CPC [122]	$X_{[1, t]}$	-	$z_{t+c}, c \geq 1$
Bidirectional CPC [123]	$X_{[1, t]}$ or $X_{[t, T]}$	-	z_{t+c} or $z_{t-c}, c \geq 1$
vq-wav2vec [124]	$X_{[1, t]}$	-	$q_{t+c}, c \geq 1$
wav2vec 2.0 [125], wav2vec-C [126] ²	X	$Z_{-[t, t+k]}$	$Q_{[t, t+k]}$
w2v-BERT [127]	X	$Z_{-[t, t+k]}$	$Q_{[t, t+k]}$ and $C_{[t, t+k]}$
Speech SimCLR [128] ³	AUGMENT ₁ (X) and AUGMENT ₂ (X)		AUGMENT ₂ (Z) and AUGMENT ₁ (Z)
PREDICTIVE MODELS			
Discrete BERT [124], [129] ⁴	$C_{-[t, t+k]}$		$C_{[t, t+k]}$
HuBERT [130] ⁵ , WavLM [131] ⁶	X	$Z_{-[t, t+k]}$	$C_{[t, t+k]}$
data2vec [132]	X	$Z_{-[t, t+k]}$	$\sum_l \bar{H}_{[t, t+k]}^{(l)}$
BEST-RQ [133] ⁷	$X_{-[t, t+k]}$		$C_{[t, t+k]}$

Japanese speech SSL (search hugging face)

- rinna/japanese-hubert-large
 - trained on ReasonSpeech (TV data)
- imprt/kushinada-hubert-large, imprt/izanami-wav2vec2-large
 - trained on TV data
- sarulab-speech/hubert-base-jtube
 - trained on YouTube data

Conclusion / まとめ

Conclusion

- Self-supervised learning / 自己教師あり学習
 - Learn to represent features from unlabeled data.
- Contrastive / 対照型 → learn to discriminative features.
- Predictive / 予測型 → learn to semantic features.
- Generative / 生成型 → learn to acoustic features.

Next: guest talk by Yuki Okamoto / 岡本 悠希, Ph.D

Project Research Associate at
UTokyo
Environmental sound synthesis
and conversion

